

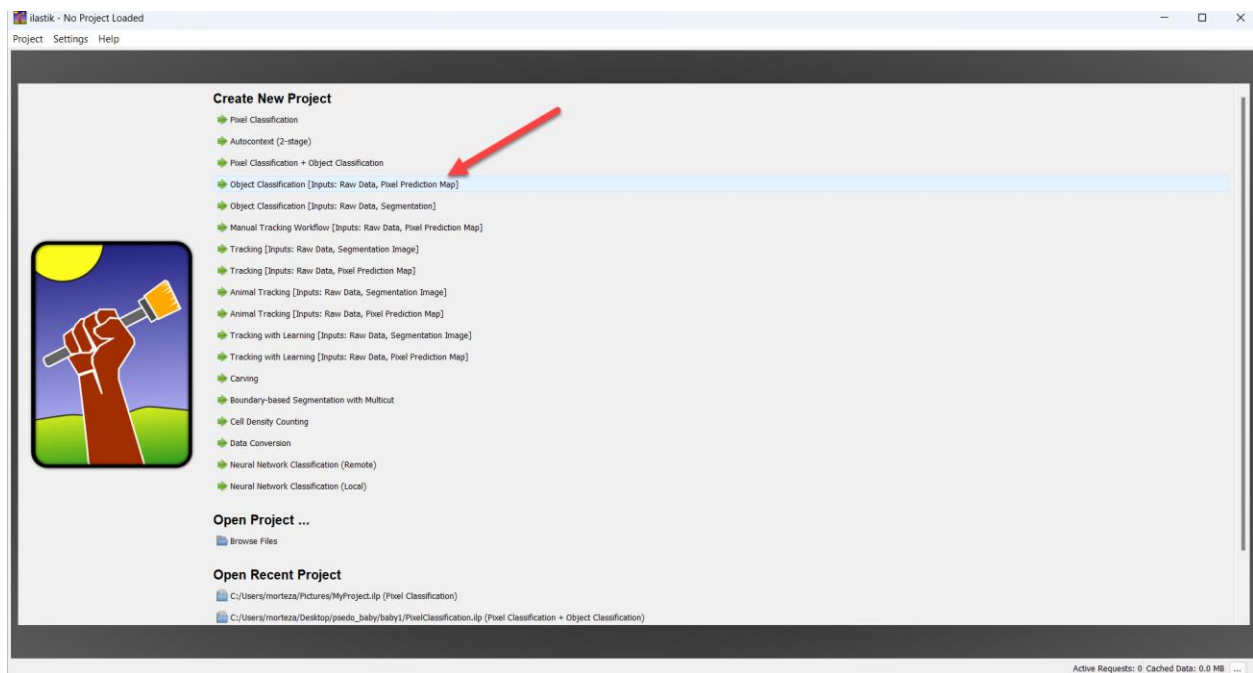
## Object Classification workflow:

“As the name suggests, the object classification workflow aims to classify full objects, based on object-level features and user annotations. An object in this context is a set of pixels that belong to the same instance. In order to do so, the workflow needs segmentation images besides the usual raw image data, that can be generated with the Pixel Classification Workflow.” (Ilastik website)

Here is my workflow:

### 1. Create New Project

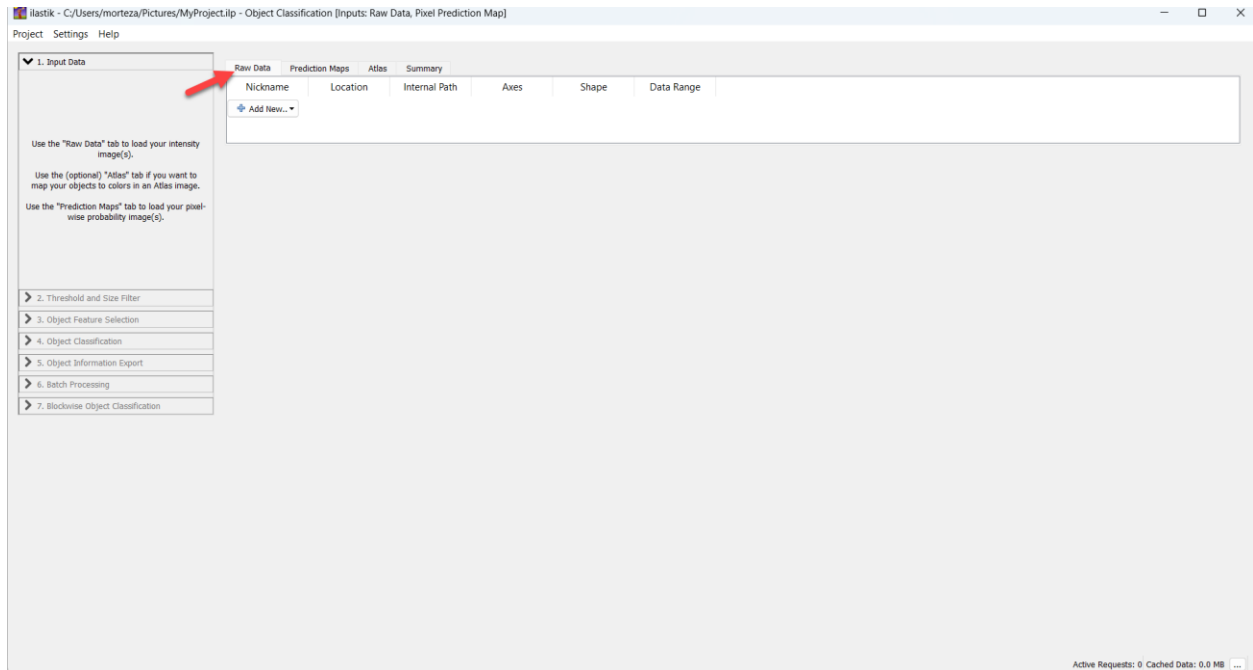
Click on **Object Classification [Input: Raw Data, Pixel Prediction Map]**



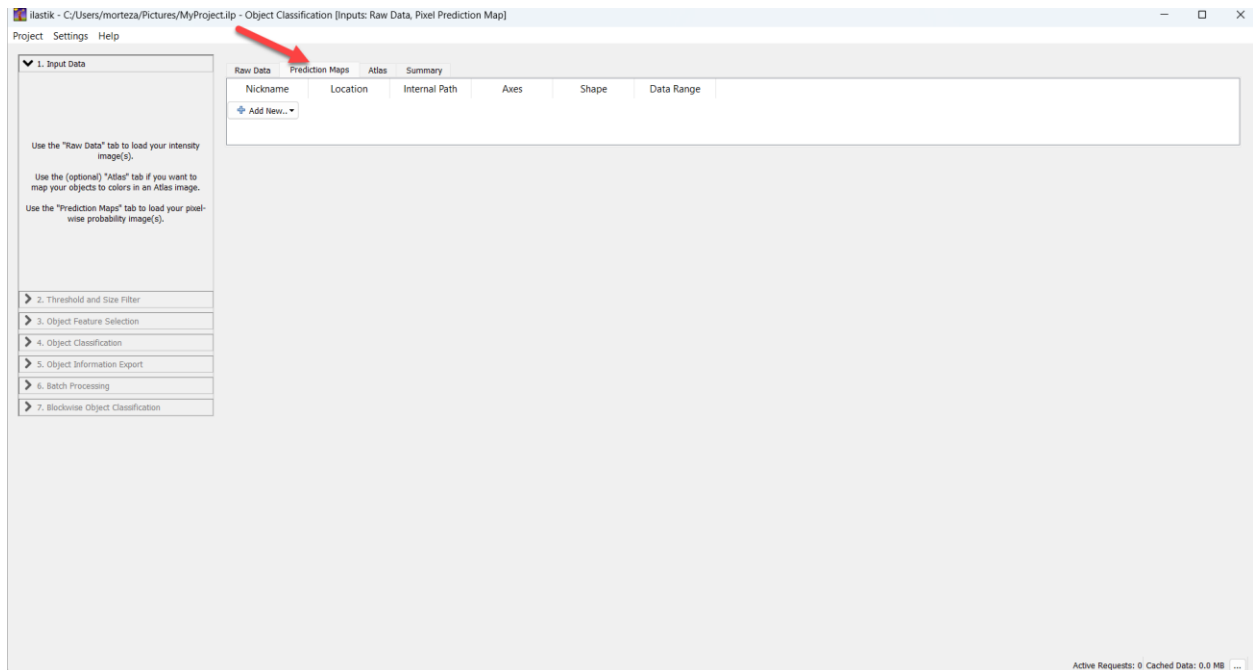
### 2. Input Data

The data input applet of this workflow expects you to load the probability maps in addition to the raw data:

a. Click on the 'Raw Data' tab and import raw images:

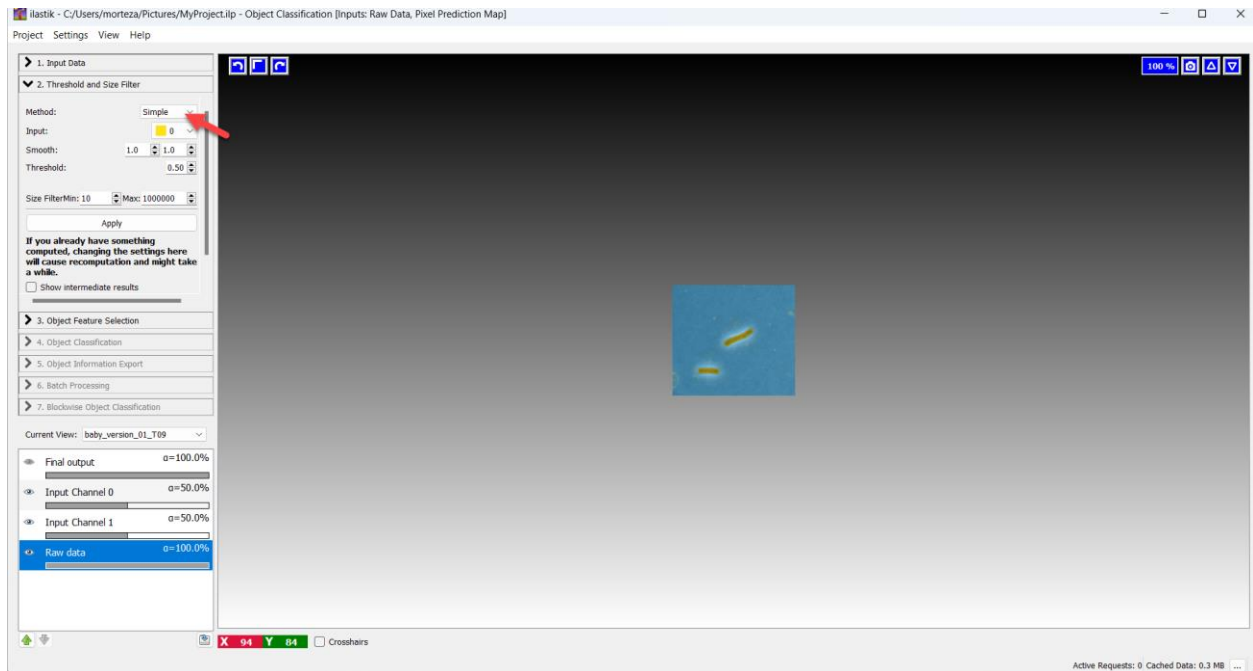


b. Click on the 'Prediction Maps' tab and import probability maps (output of pixel classification workflow):

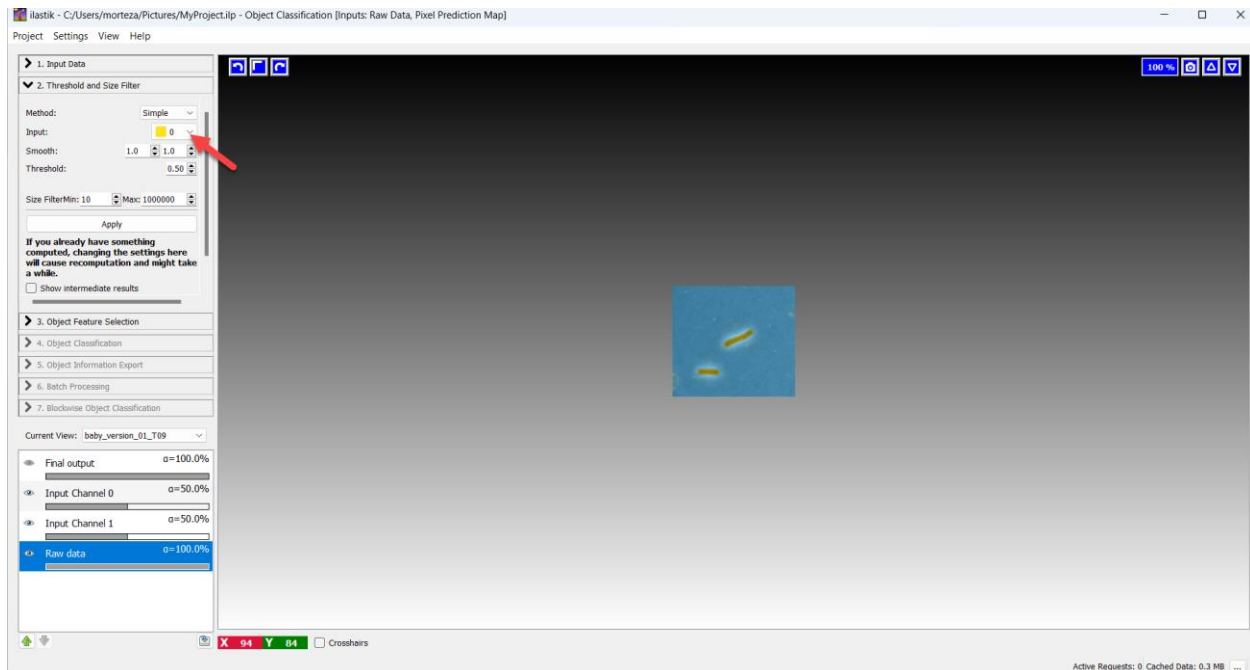


### 3. threshold your data

I select the **Simple** algorithm to threshold my data



To start, I select the yellow color channel as my input by choosing it from the dropdown menu in the "Input" section. This is because I use the yellow color to colorize pixels during pixel classification.



After selecting the yellow color channel as the input, the next step is to adjust the settings for smoothing, thresholding, and size filtering in Ilastik. Firstly, I set the value of the **Smooth** parameter to 1 for X and Y dimensions.

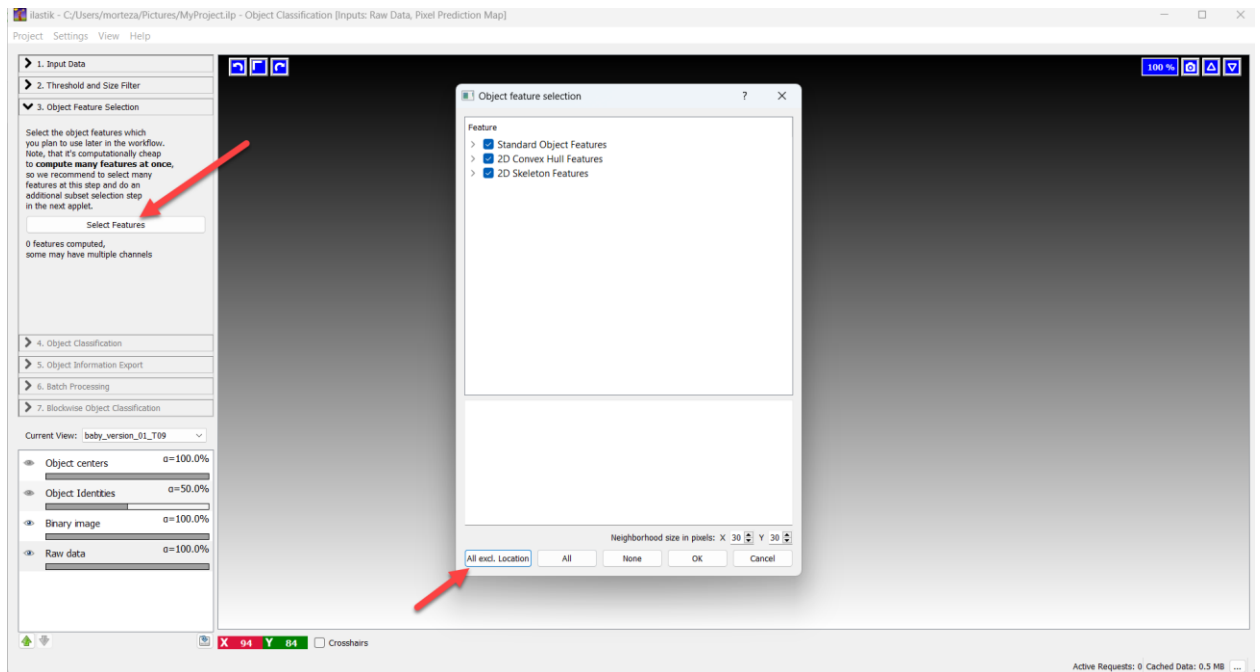
Next, I set the **Threshold** value to 0.5, which determines the probability threshold above which a pixel is classified as belonging to the yellow object. However, in some cases where Ilastik detects two closely located objects as one object, I adjust the threshold value to a range of 0.55 to 0.63 to ensure accurate separation.

Finally, I configure the **Size Filter** by setting the minimum and maximum values to 10 and 1,000,000 pixels respectively. This filter removes small or excessively large objects from the final classification result, allowing me to focus on objects within a desired size range.

Once all the settings are adjusted, I apply this configuration to my images, allowing Ilastik to process and classify the pixels based on the specified parameters.

#### 4. Object Feature Selection:

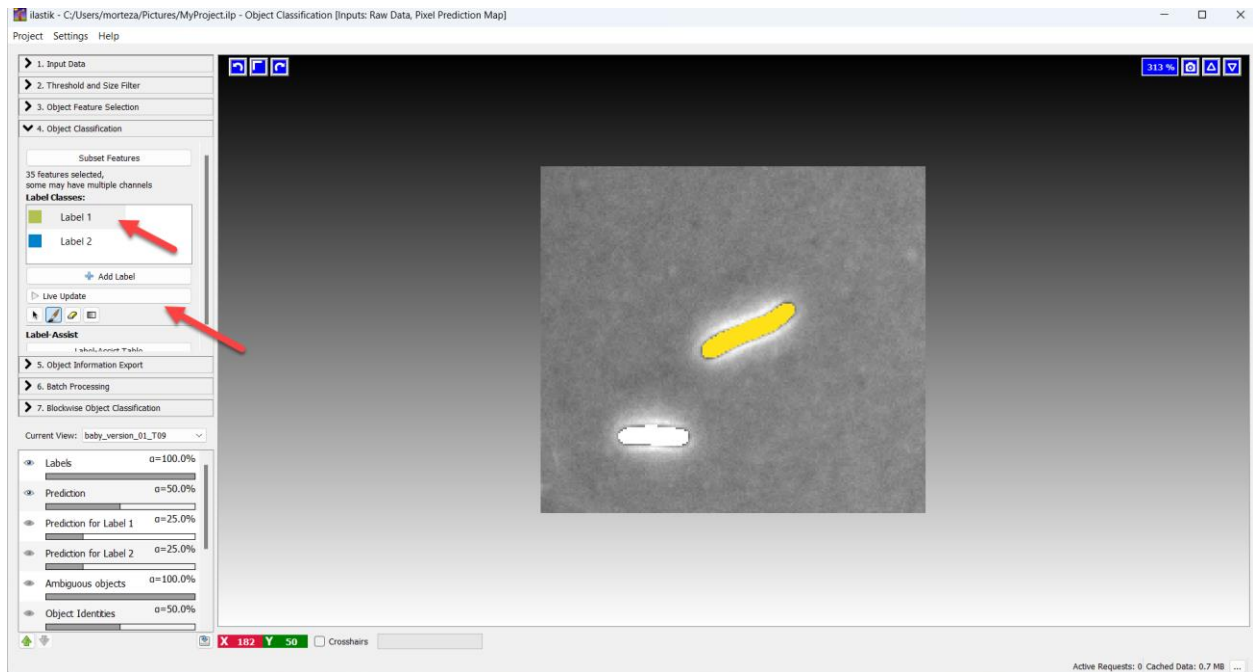
I click on the **Select Features** button to bring up a feature selection dialog. I selected **All excl. Location**.



## 5. Object Classification

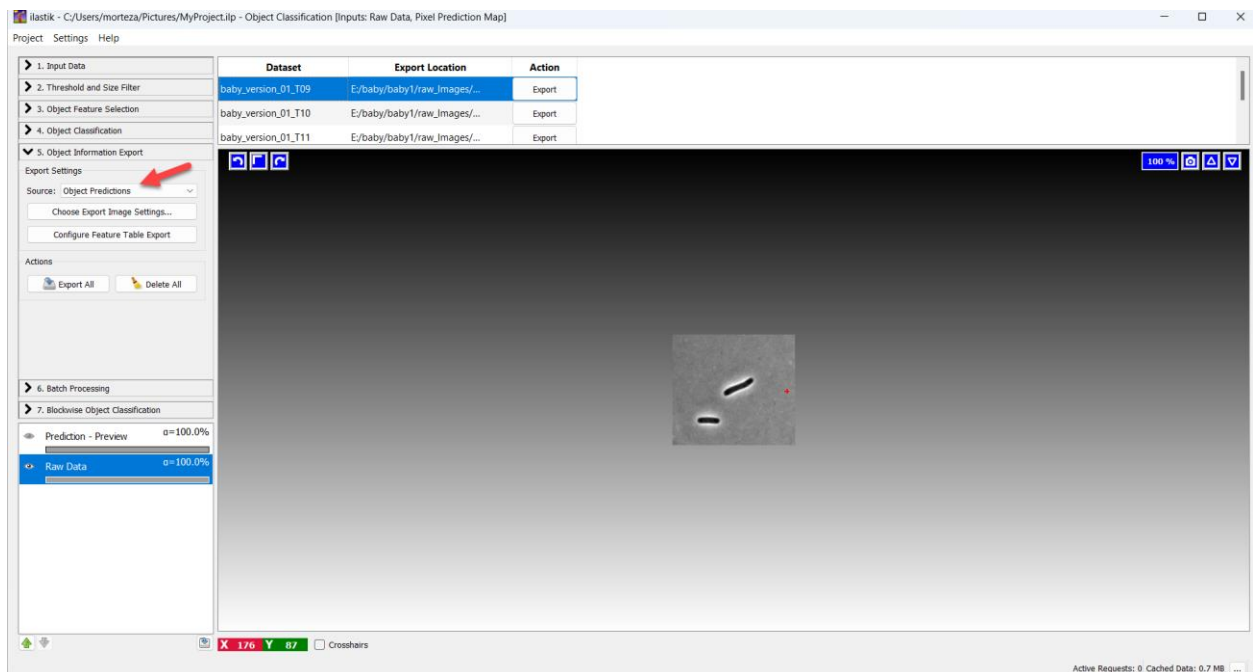
After choosing the yellow color label, I proceed to apply it to one of the bacteria by coloring it accordingly. This labeling process enables Ilastik to recognize and distinguish the bacteria as a specific object class.

Subsequently, I click on the "Live Update" button to initiate the process.

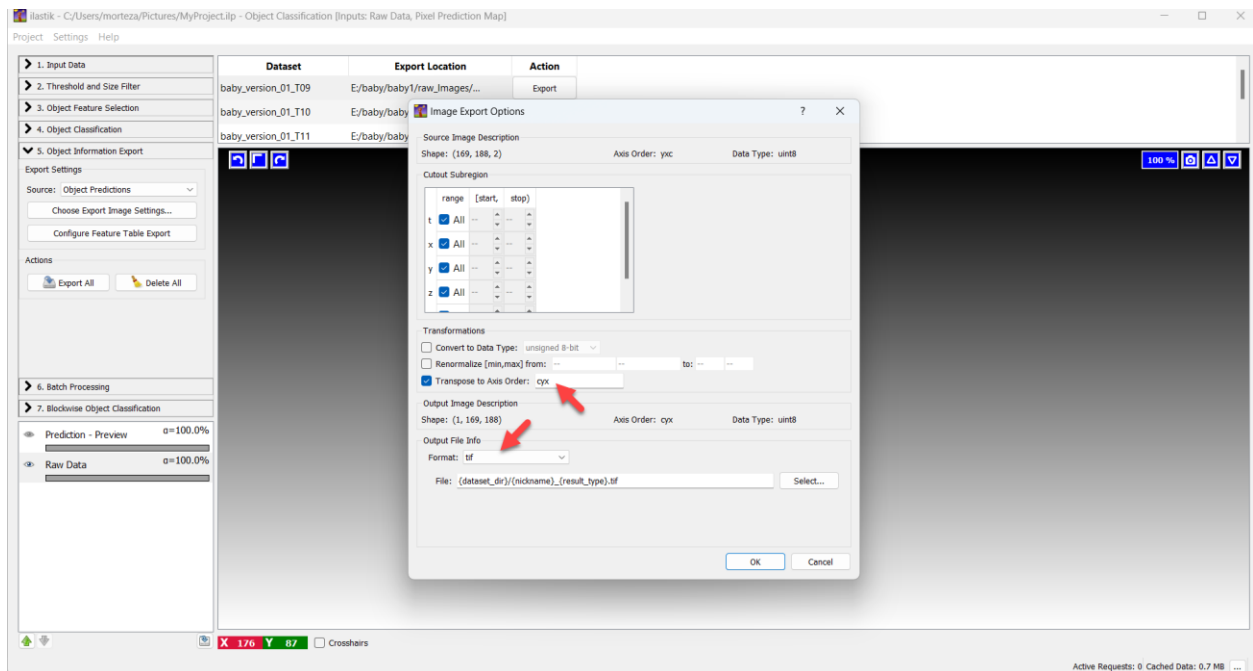


## 6. Object Information Export

I choose 'Object Predictions' as the source of the output.



Then I click on 'Choose Export Image Settings...' and define 'cyx' as the axis order and 'tif' as the desired output format.



Lastly, I click on the "Export All" option to export the images.

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